

Lecture 8:

Statistics for Data Science



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**Numerical Methods for
Computational Science**

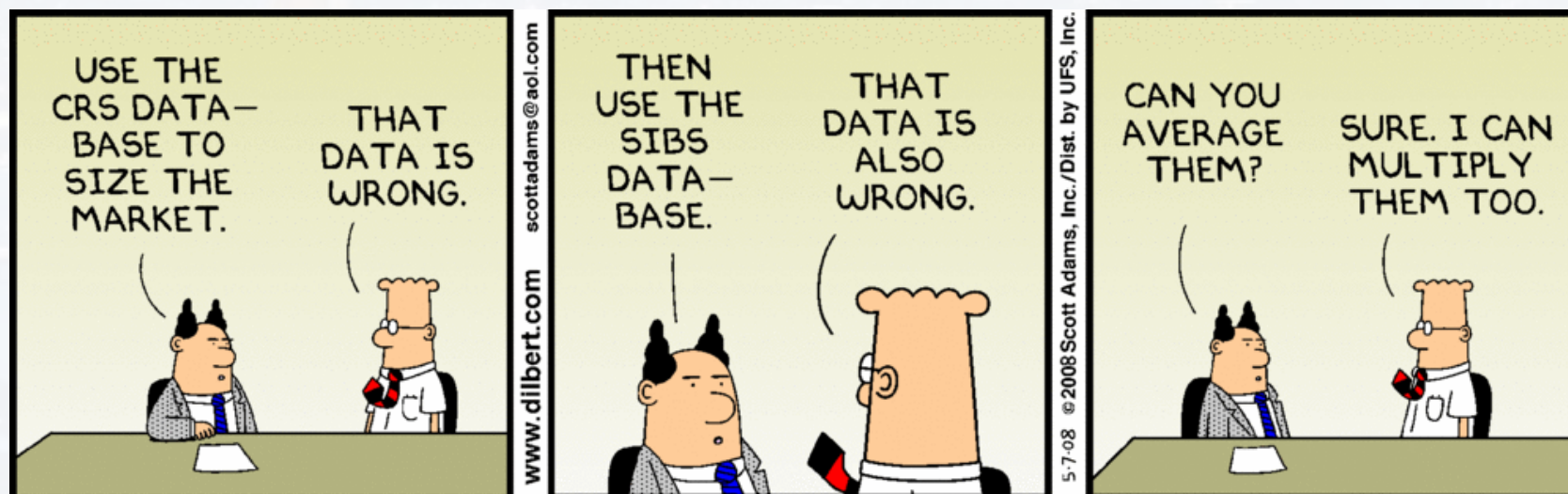
Course Map

Week 1:	Introduction to Scientific Computing and Python Libraries
Week 2:	Linear Algebra Fundamentals
Week 3:	Vector Calculus
Week 4:	Numerical Differentiation and Integration
Week 5:	Solving Nonlinear Equations
Week 6:	Probability Theory Basics
Week 7:	Random Variables and Distributions
Week 8:	Statistics for Data Science
Week 9:	Eigenvalues and Eigenvectors
Week 10:	Simulation and Monte Carlo Method
Week 11:	Data Fitting and Regression
Week 12:	Optimization Techniques
Week 13:	Machine Learning Fundamentals



Outline

- Expectation, Variance, and Covariance
- Hypothesis Testing
- Confidence Intervals



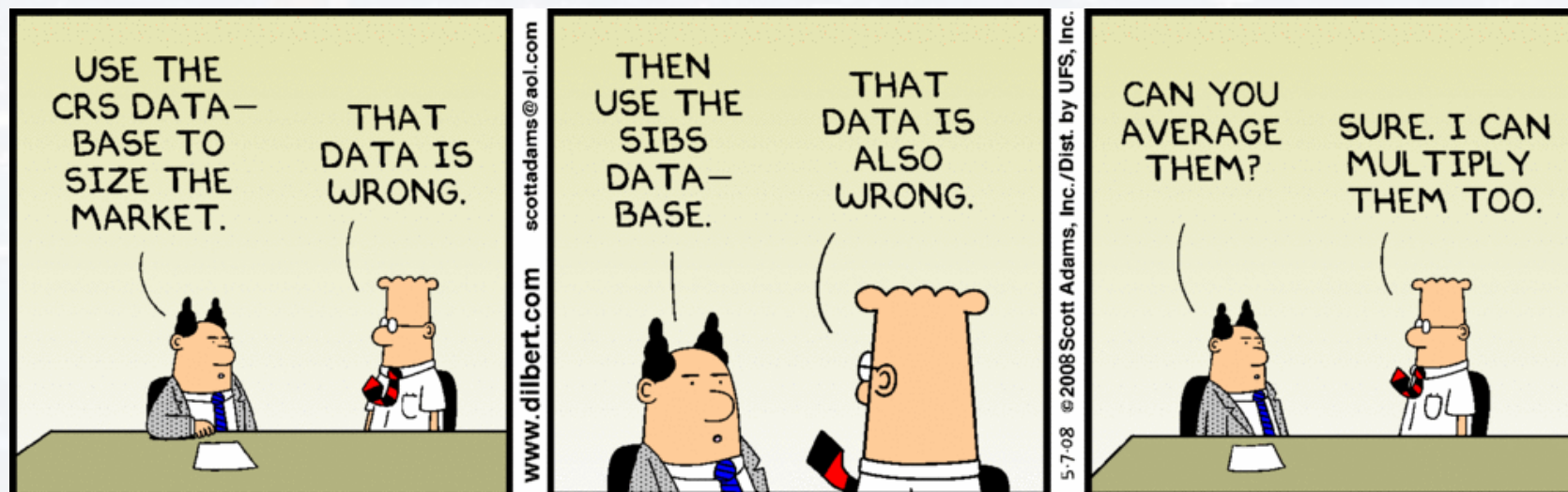


Outline

- Expectation, Variance, and Covariance

- Hypothesis Testing

- Confidence Intervals





the mean μ

(barycenter)

the variance σ^2

(natural scatter)

discrete (= countable)

$$\mu = E(x) = \sum_i x_i p(x_i)$$

$$\sigma^2 = \text{var}(x) = \sum_i (x_i - \mu)^2 p(x_i)$$

continuous

$$\mu = E(x) = \int x p(x) dx$$

$$\sigma^2 = \text{var}(x) = \int (x - \mu)^2 p(x) dx$$



$$\mu = E(x) = \int x p(x) dx$$

$$\sigma^2 = \text{var}(x) = \int (x - \mu)^2 p(x) dx$$

$$\text{var}(x) = \int (x - \mu)^2 p(x) dx = E([x - \mu]^2)$$

variance can be interpreted as **mean of $[x - \mu]^2$**

$$= E(x^2 - 2x\mu + \mu^2)$$

$$= \int [x^2 - 2x\mu + \mu^2] p(x) dx$$

$$= \int x^2 p(x) dx - 2\mu \int x p(x) dx + \mu^2 \int p(x) dx$$

$$= E(x^2) - 2\mu E(x) + \mu^2 E(1)$$

2nd axiom $\int p(x) dx = 1$

$$= E(x^2) - 2\mu E(x) + \mu^2$$

$$\mu = E(x)$$

$$\sigma^2 = E(x^2) - E(x)^2$$



$$\sigma^2 = E(x^2) - E(x)^2$$

plotting two sets of random number: x_1 and x_2

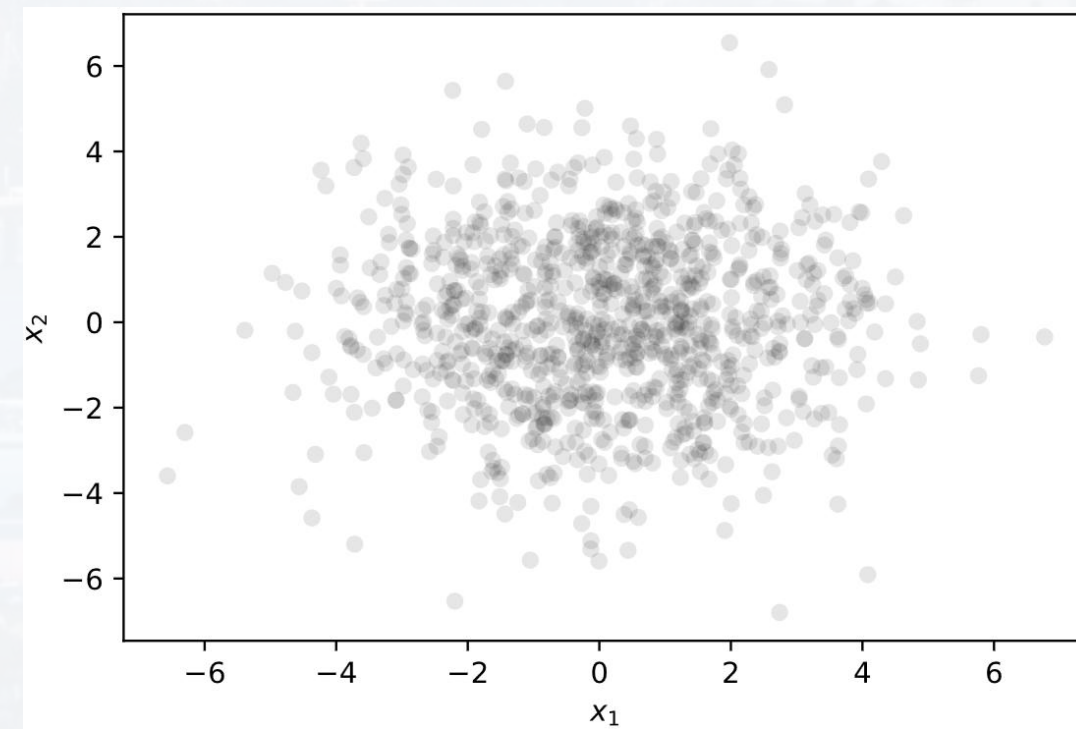
```
x1 = np.random.normal(0,2,(1000,))  
x2 = np.random.normal(0,2,(1000,))
```

```
plt.scatter(x1, x2, color = 'k', alpha = 0.1, edgecolor = 'none')  
plt.xlabel('$x_1$')  
plt.ylabel('$x_2$')
```

x_1 and x_2 are unrelated and
mutually **independent**
→ featureless data cloud

$$\mu = E(x) = \int x p(x) dx$$

$$\sigma^2 = var(x) = \int (x - \mu)^2 p(x) dx$$





$$\sigma^2 = E(x^2) - E(x)^2$$

plotting two sets of random number: x_1 and x_2

```
x1 = np.random.normal(0, 2, (1000,))
```

```
x2 = np.random.normal(0, 20, (1000,))
```

```
plt.scatter(x1, x2, color = 'k', alpha = 0.1, edgecolor = 'none')
```

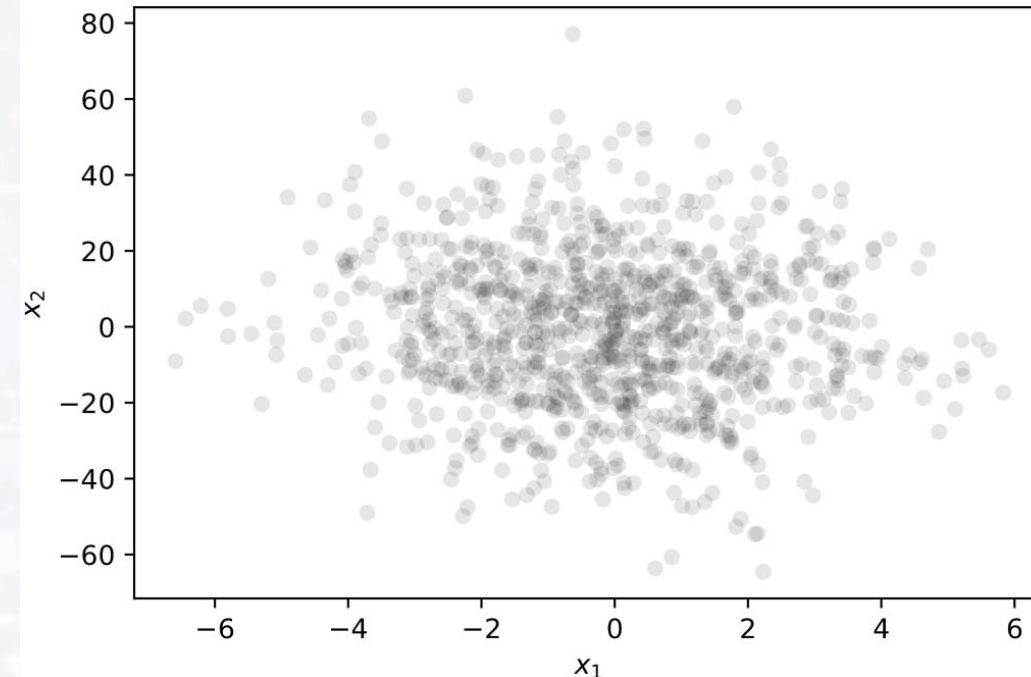
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plt.xlabel('$x_1$')
```

```
plt.ylabel('$x_2$')
```

x_1 and x_2 are unrelated and
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→ featureless data cloud

$$\mu = E(x) = \int x p(x) dx$$

$$\sigma^2 = var(x) = \int (x - \mu)^2 p(x) dx$$





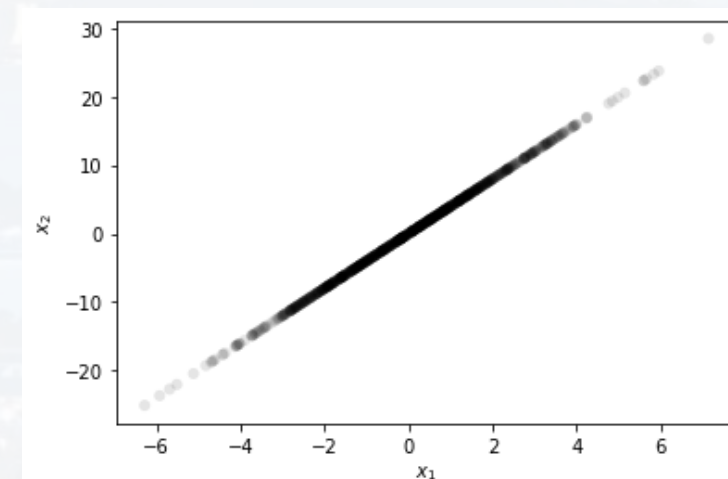
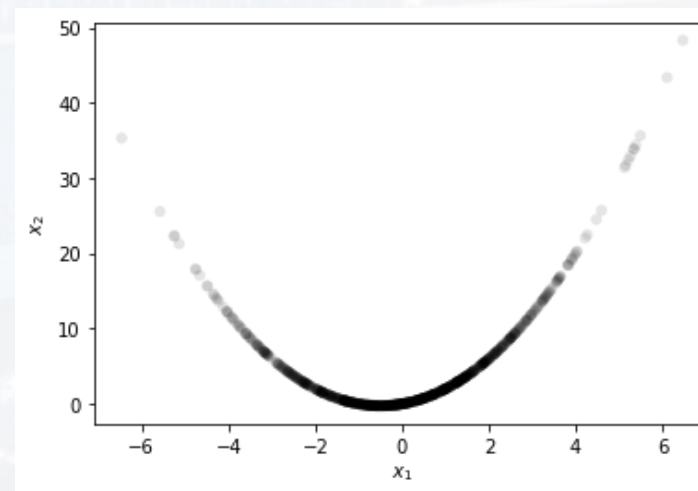
$$\sigma^2 = E(x^2) - E(x)^2$$

plotting two sets of random number: x_1 and x_2

```
x1 = np.random.normal(0,2,(1000,))  
x2 = x1**2 + x1  
#x2 = 4*x1
```

```
plt.scatter(x1, x2, color = 'k', alpha = 0.1, edgecolor = 'none')  
plt.xlabel('$x_1$')  
plt.ylabel('$x_2$')
```

based on the shape of the
data cloud
→ prediction how x_1 and x_2
are related, i. e.
how they **correlate**



$$\mu = E(x) = \int x p(x) dx$$

$$\sigma^2 = var(x) = \int (x - \mu)^2 p(x) dx$$



$$\sigma^2 = E(x^2) - E(x)^2$$

$$\mu = E(x) = \int x p(x) dx$$

$$\sigma^2 = \text{var}(x) = \int (x - \mu)^2 p(x) dx$$

```
x1 = np.random.normal(0,2,(1000,))
```

```
x2 = np.random.normal(3,3,(1000,))
```

```
x3 = np.random.uniform(0,5,(1000,))
```

```
x4 = 5*np.random.uniform(3,4,(1000,))
```

```
x5 = np.sqrt(x4)
```

```
x6 = x1 + x2
```

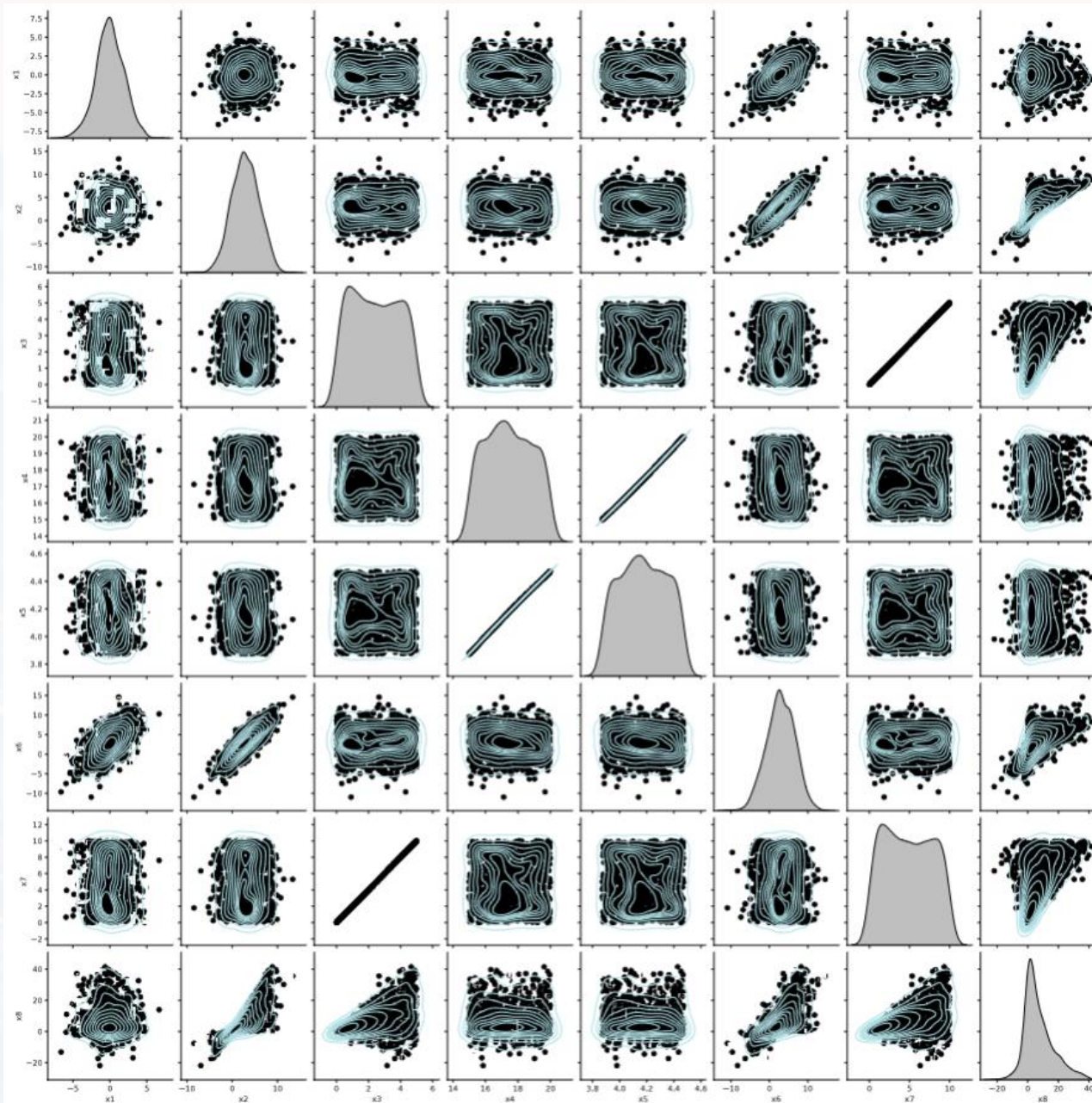
```
x7 = 2*x3
```

```
x8 = x3*x2
```

```
All = np.vstack((x1, x2, x3, x4, x5, x6, x7, x8))
```

```
data = pd.DataFrame(All.transpose(),  
                    columns = ['x1', 'x2', 'x3', 'x4', 'x5', 'x6', 'x7', 'x8'])
```

```
out = sns.pairplot(data, kind = "kde", \  
                  plot_kws = {'color':[176/255, 224/255, 230/255]}, \  
                  diag_kws = {'color':'black'})  
out.map_offdiag(plt.scatter, color = 'black')
```

$$\mu = E(x) = \int x p(x) dx$$

$$\sigma^2 = \text{var}(x) = \int (x - \mu)^2 p(x) dx$$

based on the shape of the
data cloud

→ prediction how x_1 and x_2
are related, i. e.
how they **correlate**

→ how to quantify?



a, b = const

$$\sigma^2 = E(x^2) - E(x)^2$$

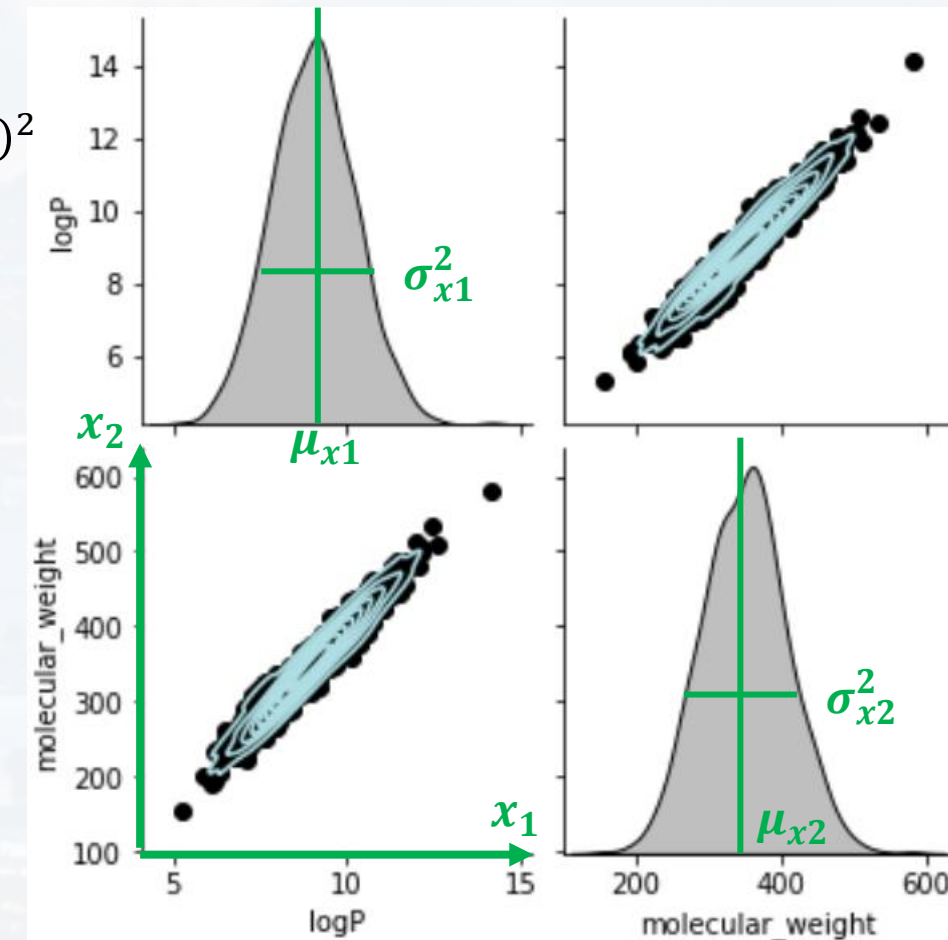
$$\mu = E(x) = \int x p(x) dx$$

$$\sigma^2 = \text{var}(x) = \int (x - \mu)^2 p(x) dx$$

$$\text{var}([a x_1 + b x_2]) = E([a x_1 + b x_2]^2) - E(a x_1 + b x_2)^2$$

$$= E(a^2 x_1^2 + 2ab x_1 x_2 + b^2 x_2^2) - E(a x_1 + b x_2)^2$$

$$= a^2 E(x_1^2) + 2ab E(x_1 x_2) + b^2 E(x_2^2) - E(a x_1 + b x_2)^2$$





a, b = const

$$\sigma^2 = E(x^2) - E(x)^2$$

$$\mu = E(x) = \int x p(x) dx$$

$$\sigma^2 = \text{var}(x) = \int (x - \mu)^2 p(x) dx$$

$$\text{var}([a x_1 + b x_2]) = E([a x_1 + b x_2]^2) - E(a x_1 + b x_2)^2$$

$$= E(a^2 x_1^2 + 2ab x_1 x_2 + b^2 x_2^2) - E(a x_1 + b x_2)^2$$

$$= a^2 E(x_1^2) + 2ab E(x_1 x_2) + b^2 E(x_2^2) - E(a x_1 + b x_2)^2$$

$$= a^2 E(x_1^2) + 2ab E(x_1 x_2) + b^2 E(x_2^2) - [aE(x_1) + bE(x_2)]^2$$

$$= a^2 E(x_1^2) - a^2 E(x_1)^2 + b^2 E(x_2^2) - b^2 E(x_2)^2 + 2ab E(x_1 x_2) - 2ab E(x_1)E(x_2)$$

$$a^2 \text{var}(x_1)$$

$$b^2 \text{var}(x_2)$$

$$2ab \text{cov}(x_1, x_2)$$

$$= a^2 \text{var}(x_1) + b^2 \text{var}(x_2) + 2ab \text{cov}(x_1, x_2)$$

$$\text{cov}(x_1, x_2) = \text{cov}(x_2, x_1) = E(x_1 x_2) - E(x_1)E(x_2)$$

covariance



$$\text{cov}(x_1, x_2) = \text{cov}(x_2, x_1) = E(x_1 x_2) - E(x_1)E(x_2)$$

What does the covariance tell us?

- 1) geometrical interpretation → next lecture
- 2) arithmetical interpretation

$$\mu = E(x) = \int x p(x) dx$$

$$\sigma^2 = \text{var}(x) = \int (x - \mu)^2 p(x) dx$$



$$\text{cov}(x_1, x_2) = \text{cov}(x_2, x_1) = E(x_1 x_2) - E(x_1)E(x_2)$$

What does the covariance tell us?

1) geometrical interpretation → next lecture

2) arithmetical interpretation

a) x_1 and x_2 are independent

$$E(x_1 x_2) - E(x_1)E(x_2)$$

$$= \iint x_1 x_2 p(x_1) p(x_2) dx_1 dx_2 - \int x_1 p(x_1) dx_1 \int x_2 p(x_2) dx_2$$

x_1 and x_2 are independent:

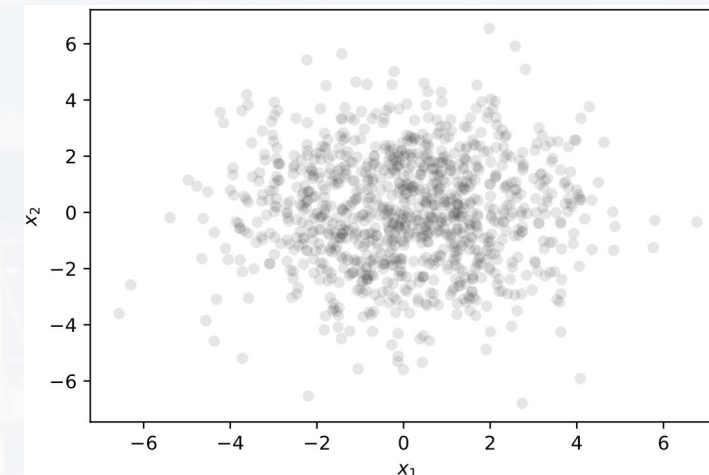
x_1 is not a function of x_2 and vice versa

x_1 cannot be predicted by x_2 and vice versa

$$= \int x_1 p(x_1) dx_1 \int x_2 p(x_2) dx_2 - \int x_1 p(x_1) dx_1 \int x_2 p(x_2) dx_2 = 0$$

$$\mu = E(x) = \int x p(x) dx$$

$$\sigma^2 = \text{var}(x) = \int (x - \mu)^2 p(x) dx$$



Covariance equals **zero**
if samples are **independent!**



$$\text{cov}(x_1, x_2) = \text{cov}(x_2, x_1) = E(x_1 x_2) - E(x_1)E(x_2)$$

What does the covariance tell us?

- 1) geometrical interpretation → next lecture
- 2) **arithmetical interpretation**

b) x_1 and x_2 are **not** independent

$$E(x_1 x_2) - E(x_1)E(x_2)$$

$$= \iint x_1 x_2 p(x_1) p(x_2) dx_1 dx_2 - \int x_1 p(x_1) dx_1 \int x_2 p(x_2) dx_2$$

x_1 and x_2 are **not** independent:

x_1 **is** a function of x_2 and vice versa

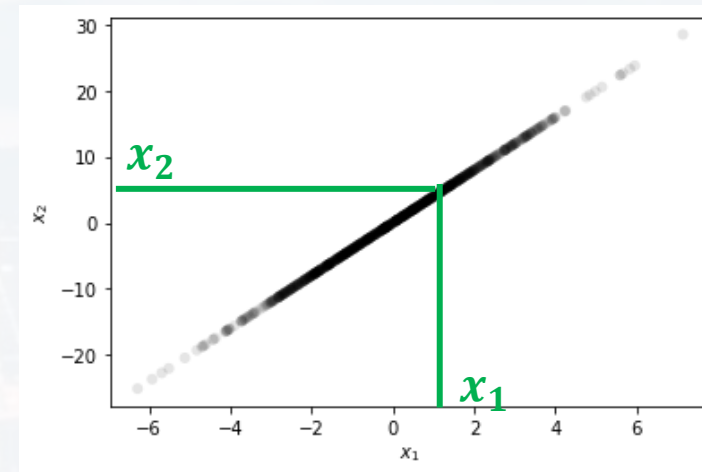
x_1 **can** be predicted by x_2 to certain degree and vice versa

$$= \iint x_1 p(x_1) x_2(x_1) p(x_2(x_1)) dx_1 dx_2(x_1) - \int x_1 p(x_1) dx_1 \int x_2 p(x_2) dx_2$$

Covariance **does not**
equal **zero**!

$$\mu = E(x) = \int x p(x) dx$$

$$\sigma^2 = \text{var}(x) = \int (x - \mu)^2 p(x) dx$$

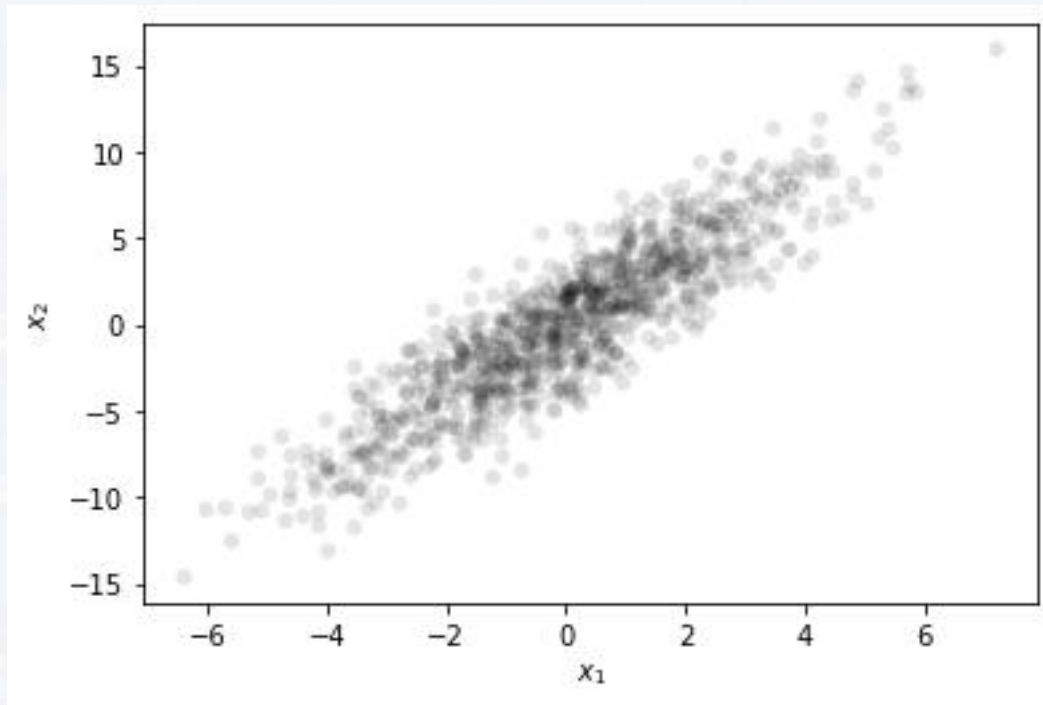




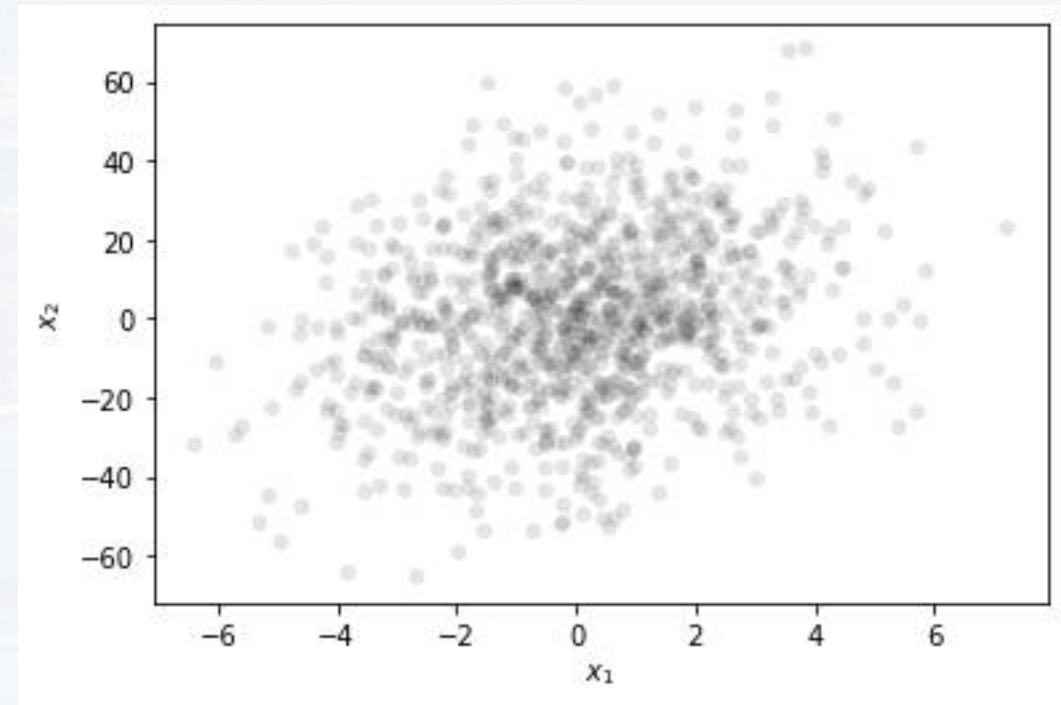
$$\text{cov}(x_1, x_2) = \text{cov}(x_2, x_1) = E(x_1 x_2) - E(x_1)E(x_2)$$

covariance

```
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x2 = 2*x1 + np.random.normal(0, 2, (1000,))
```



```
x1 = np.random.normal(0, 2, (1000,))  
x2 = 2*x1 + np.random.normal(0, 20, (1000,))
```

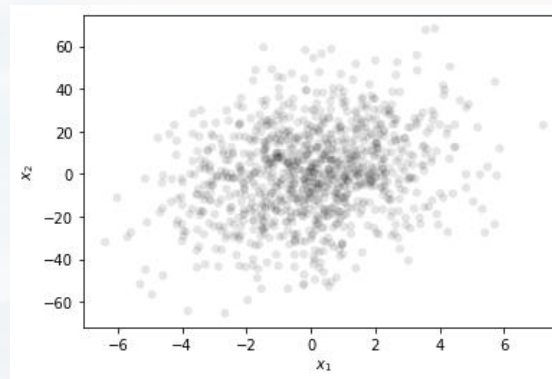
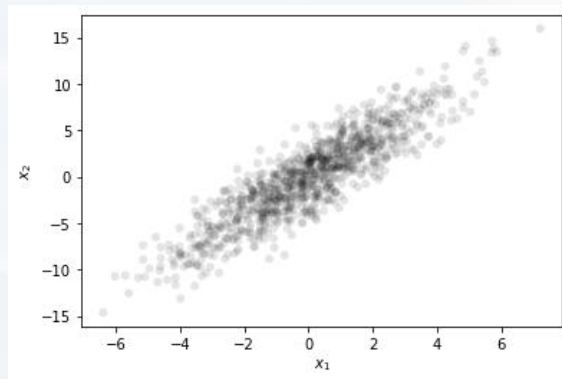


Same dependency, but different variance!



$$\text{cov}(x_1, x_2) = \text{cov}(x_2, x_1) = E(x_1 x_2) - E(x_1)E(x_2)$$

covariance



Same dependency, but different variance!

Need to scale for the variance!

Pearson's correlation
coefficient

$$\rho(x_1, x_2) = \frac{\text{cov}(x_1, x_2)}{\sqrt{\sigma_1^2 \sigma_2^2}}$$

$\rho(x_1, x_2)$:

- ranges from -1 to +1
- zero: no correlation
(completely independent)
- -1: max anti correlation
- +1: max correlation



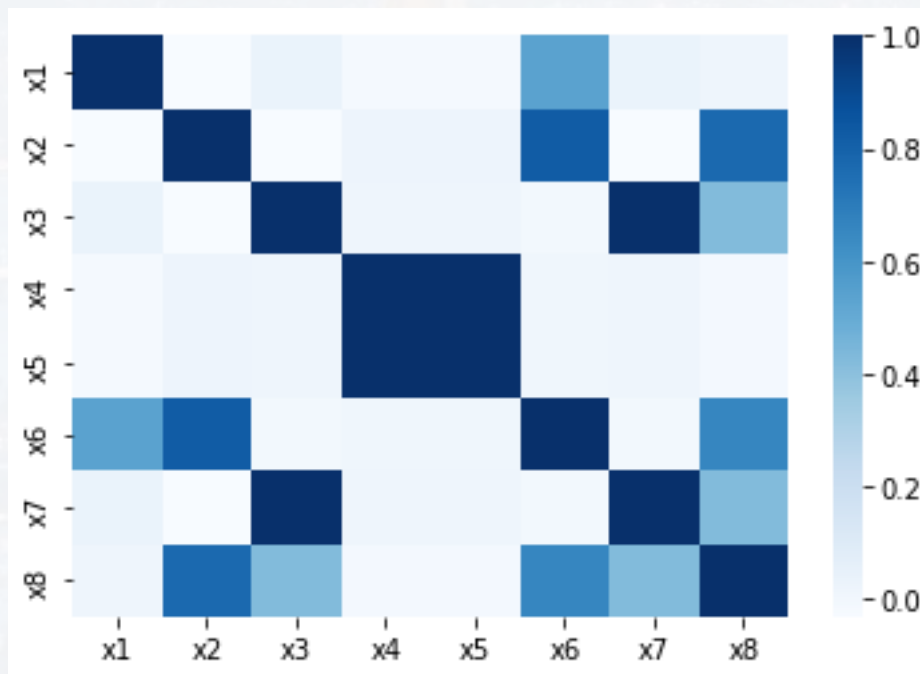
$$\text{cov}(x_1, x_2) = \text{cov}(x_2, x_1) = E(x_1 x_2) - E(x_1)E(x_2)$$

covariance

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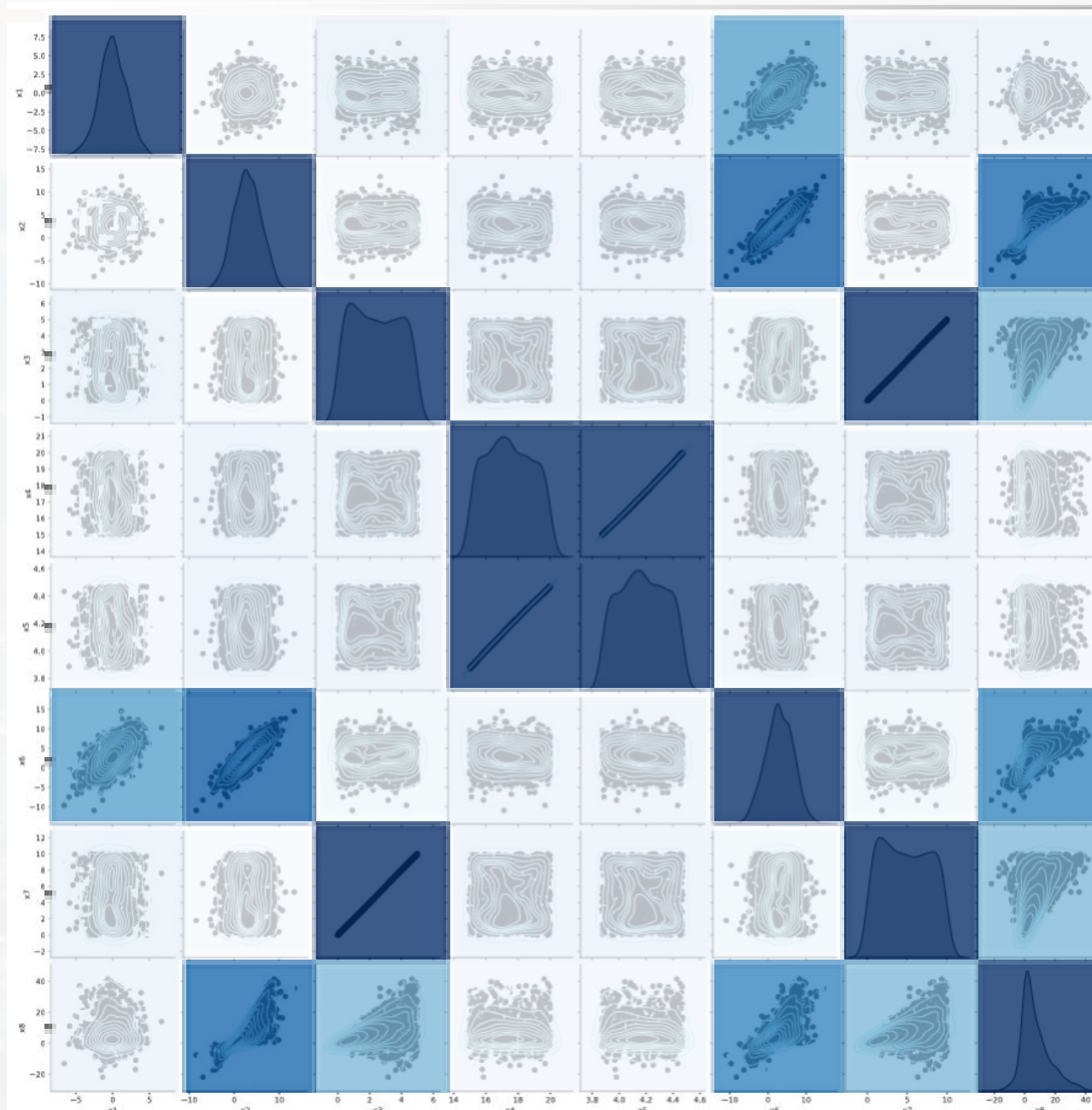
Pearson's correlation
coefficient

```
sns.heatmap(data.corr(), cmap = "Blues")
```



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$\rho(x_1, x_2)$:

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- +1: max correlation



Important quantities you should know:

mean

$$\mu = E(x) = \int x p(x) dx$$

median m

$$\int_a^m p(x) dx = \frac{1}{2}$$

variance

$$\sigma^2 = \text{var}(x) = \int (x - \mu)^2 p(x) dx$$

$$\sigma^2 = E(x^2) - E(x)^2$$

$$\sigma_{tot}^2 = \sigma_1^2 + \sigma_2^2 + 2 \text{cov}(x_1, x_2)$$

covariance

$$\text{cov}(x_1, x_2) = E(x_1 x_2) - E(x_1)E(x_2)$$

**correlation
coefficient**

$$\rho(x_1, x_2) = \frac{\text{cov}(x_1, x_2)}{\sqrt{\sigma_1^2 \sigma_2^2}}$$

note:

$$\int (x - \mu)^n p(x) dx$$

called n -th *moment*
of a pdf

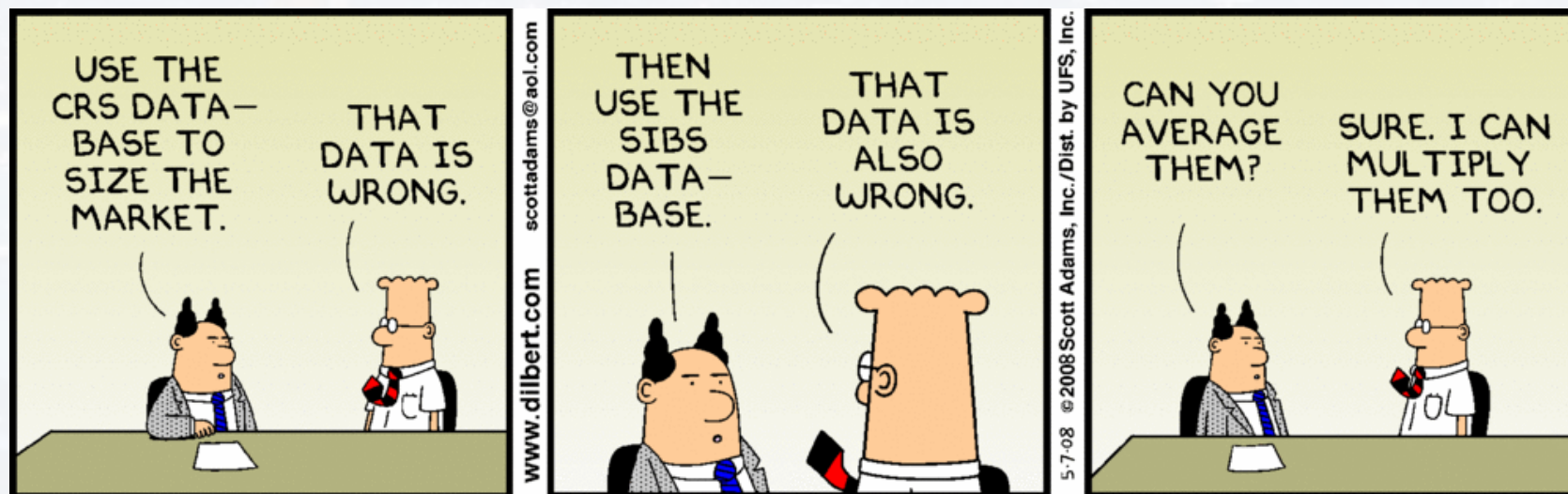


Outline

- Expectation, Variance, and Covariance

- **Hypothesis Testing**

- Confidence Intervals



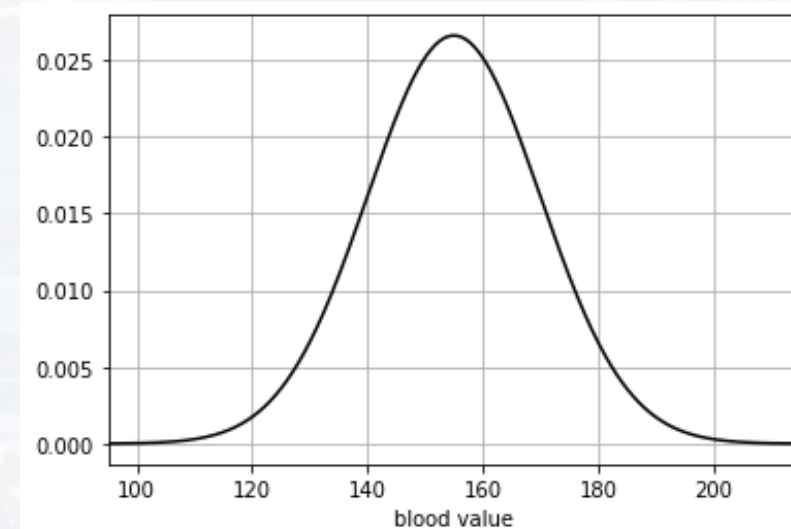


standard frequentist way (not Bayesian):

- 1) assume a **likelihood function L as your model**, aka **null hypothesis H_0**
- 2) take a datapoint x_0
- 3) calculate the **probability P** given L i. e. **given H_0 is true, that x_0 has this value or a more extreme value**
- 4) accepting or rejecting H_0 based on P and the threshold α

example:

1) a healthy person has a blood value that follows a **normal distribution** with $\mu = 155$, $\sigma = 15$, i. e. **$H_0: N(\mu = 155, \sigma = 15)$**





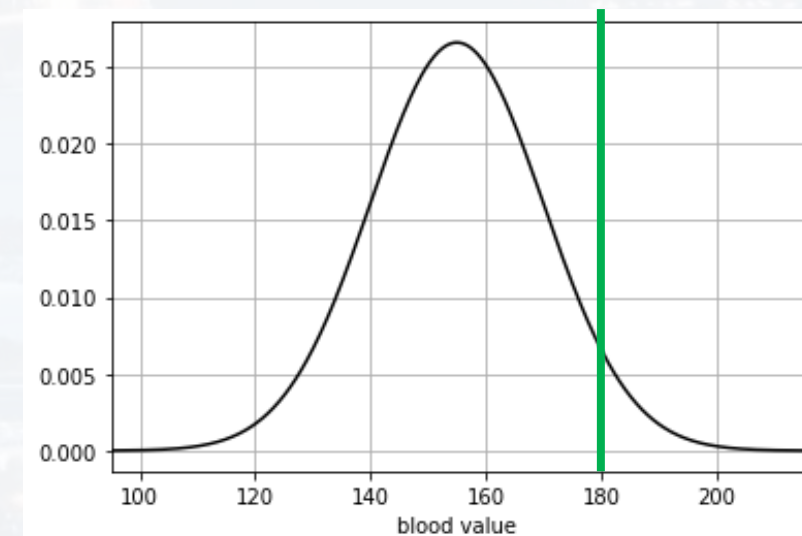
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1) a healthy person has a blood value that follows a **normal distribution** with $\mu = 155, \sigma = 15$, i. e. **$H_0: N(\mu = 155, \sigma = 15)$**

2) a patient has the value **$x_0 = 180$**





standard frequentist way (not Bayesian):

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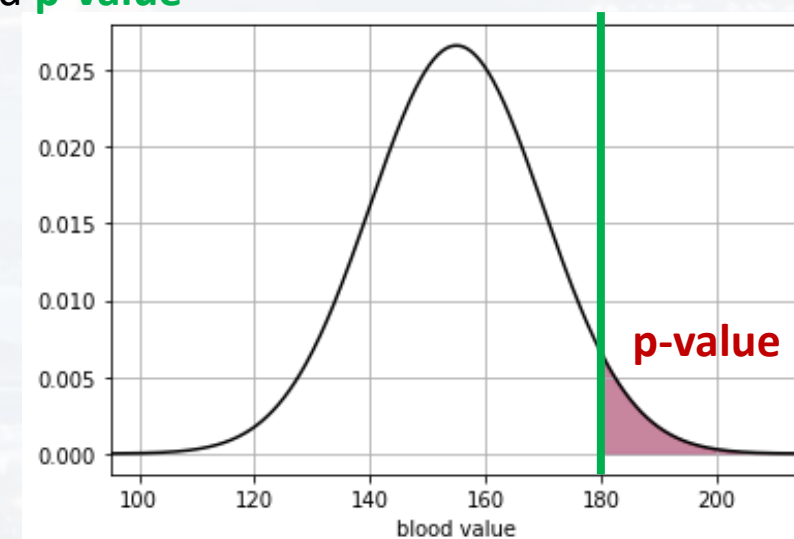
example:

1) a healthy person has a blood value that follows a **normal distribution** with

$\mu = 155, \sigma = 15$, i. e. **$H_0: N(\mu = 155, \sigma = 15)$**

2) a patient has the value **$x_0 = 180$**

3) probability **$P(x \geq 180 | H_0) = 0.048$** called **p-value**





standard frequentist way (not Bayesian):

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example:

1) a healthy person has a blood value that follows a **normal distribution** with

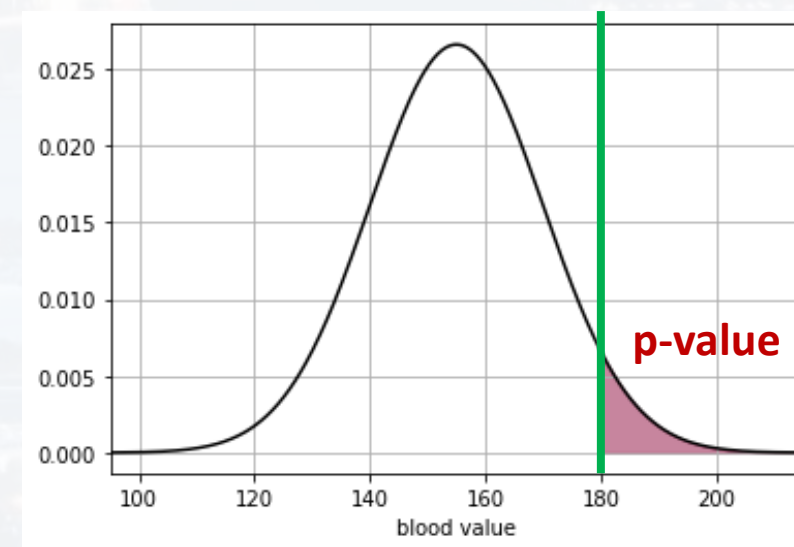
$\mu = 155, \sigma = 15$, i. e. **$H_0: N(\mu = 155, \sigma = 15)$**

2) a patient has the value **$x_0 = 180$**

3) probability **$P(x \geq 180 | H_0) = 0.048$** called **p-value**

4) for **$\alpha = 0.05$** , H_0 is **rejected**, i. e. patient is **not healthy**

→ alternative hypothesis **H_1 : not healthy**





example:

1) a healthy person has a blood value that follows a **normal distribution** with

$\mu = 155, \sigma = 15$, i. e. **$H_0: N(\mu = 155, \sigma = 15)$**

2) a patient has the value **$x_0 = 180$**

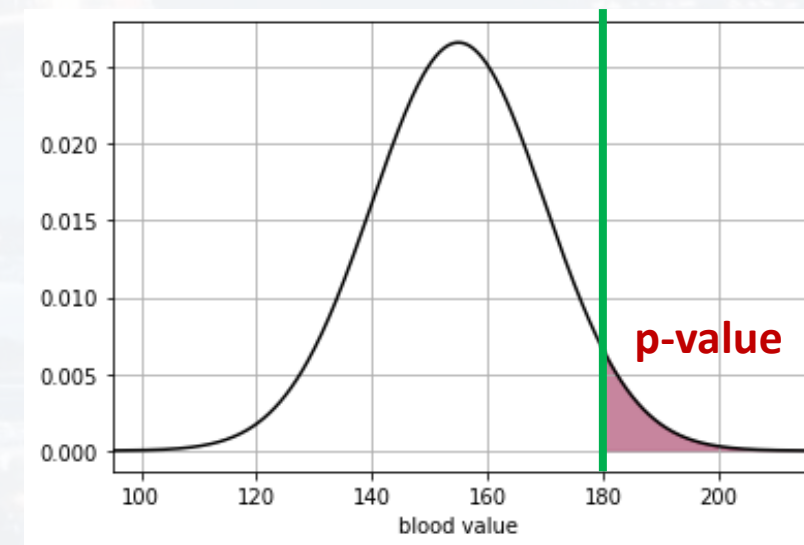
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4) for **$\alpha = 0.05$** , H_0 is **rejected**, i. e. patient is

not healthy

→ alternative hypothesis **H_1 : not healthy**

We just performed a so-called Z – Test
(comparing one value to a normal distribution)





standard frequentist way (not Bayesian):

- 1) assume a **likelihood function L as your model**, aka **null hypothesis H_0**
- 2) take a datapoint x_0
- 3) calculate the **probability P** given L i. e. **given H_0 is true, that x_0 has this value or a more extreme value**
- 4) accepting or rejecting H_0 based on P and the threshold α

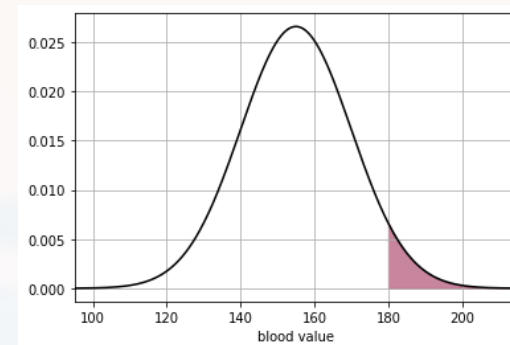
caveats:

- the model L for H_0 has to **be known**
- the p-value $P(x \geq x_0 | H_0)$ **does not** tell if H_0 is true or not
- the p-value $P(x \geq x_0 | H_0)$ **does not** tell which hypothesis is more likely
- the p-value just gives $P(x \geq x_0 | H_0)$
- the threshold α for accepting/rejecting H_0 is **arbitrary**
- we are aiming on disproving a hypothesis, by assuming it is true,
without leading to a contradiction



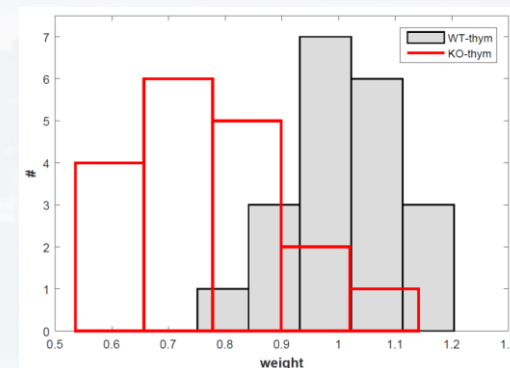
- data point x versus **normal**
- $H_0: x \in N(\mu, \sigma)$

Z-test



- **normal** versus another **normal**
- two samples of sizes n_1 and n_2

t-test

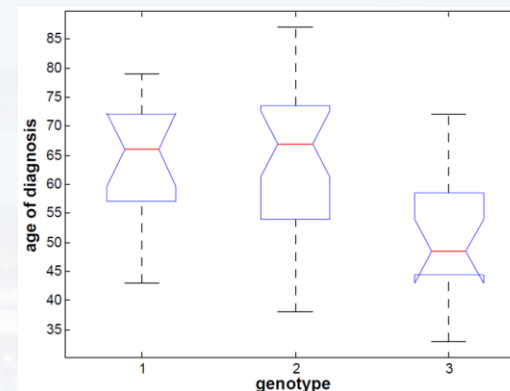


- two tail $H_0: \mu_1 = \mu_2, H_1: \mu_1 \neq \mu_2$
- right tail $H_0: \mu_1 < \mu_2, H_1: \mu_1 > \mu_2$
- left tail $H_0: \mu_1 > \mu_2, H_1: \mu_1 < \mu_2$

- **N normal dist.**

ANalysis Of VAriance

- N samples of sizes n_i
- $H_0: \mu_i = \mu_j \forall i, j, H_1: \mu_i \neq \mu_j$ for at least one pair i, j

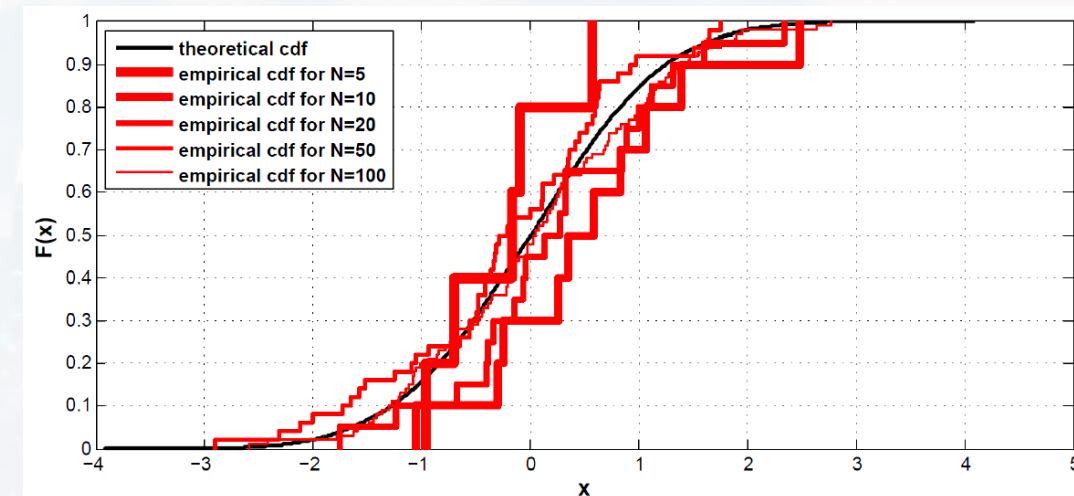




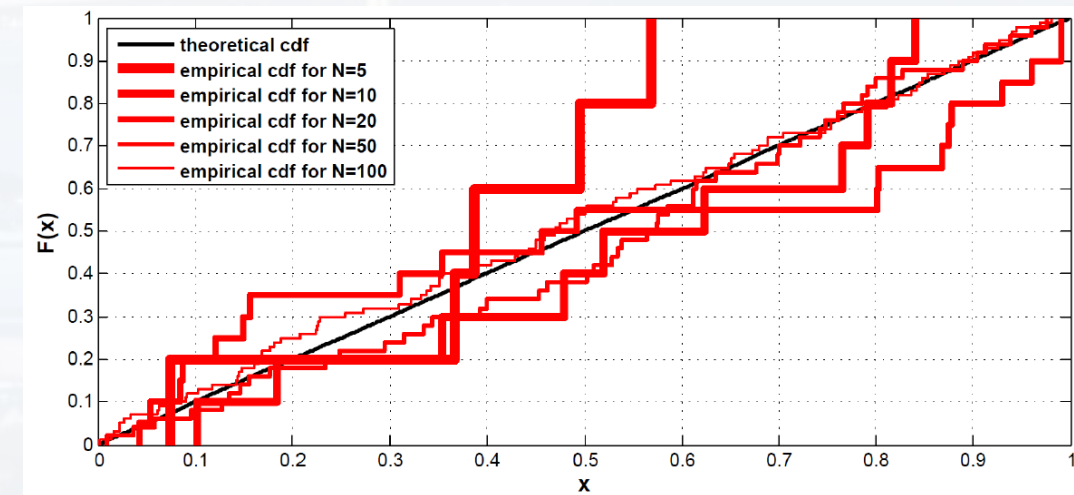
testing if a data set follows a particular distribution

Kolmogorov – Smirnov – test (KS test)

example: normal distribution:



example: uniform distribution:





testing if a data set follows a particular distribution

ranking tests (Wilcoxon)

...and many more...

They all generate a test statistic:

Z-value, t-value, F-value etc

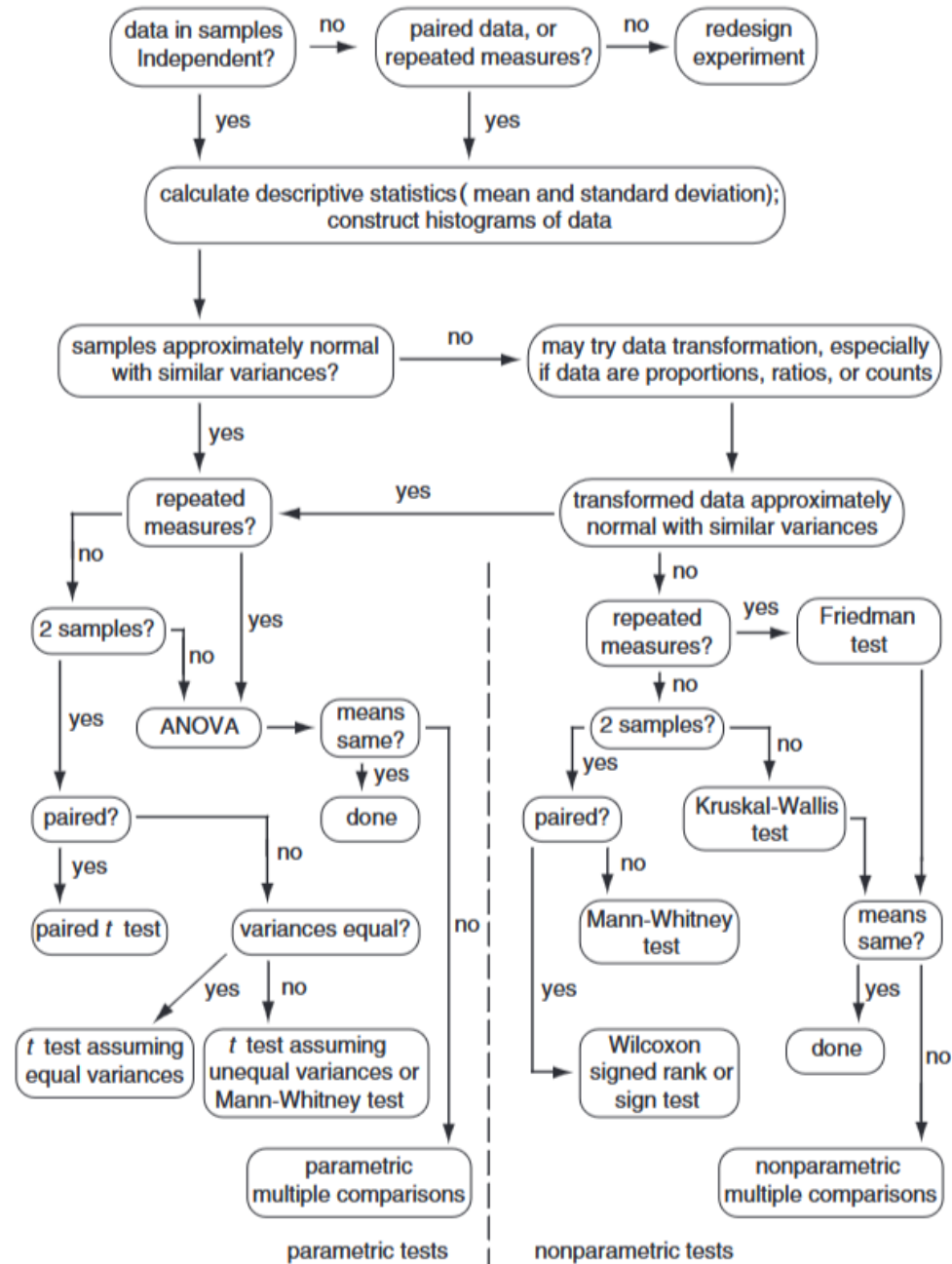
→ from that: calculating a p-value



testing if a data s

...and many more

They all generate



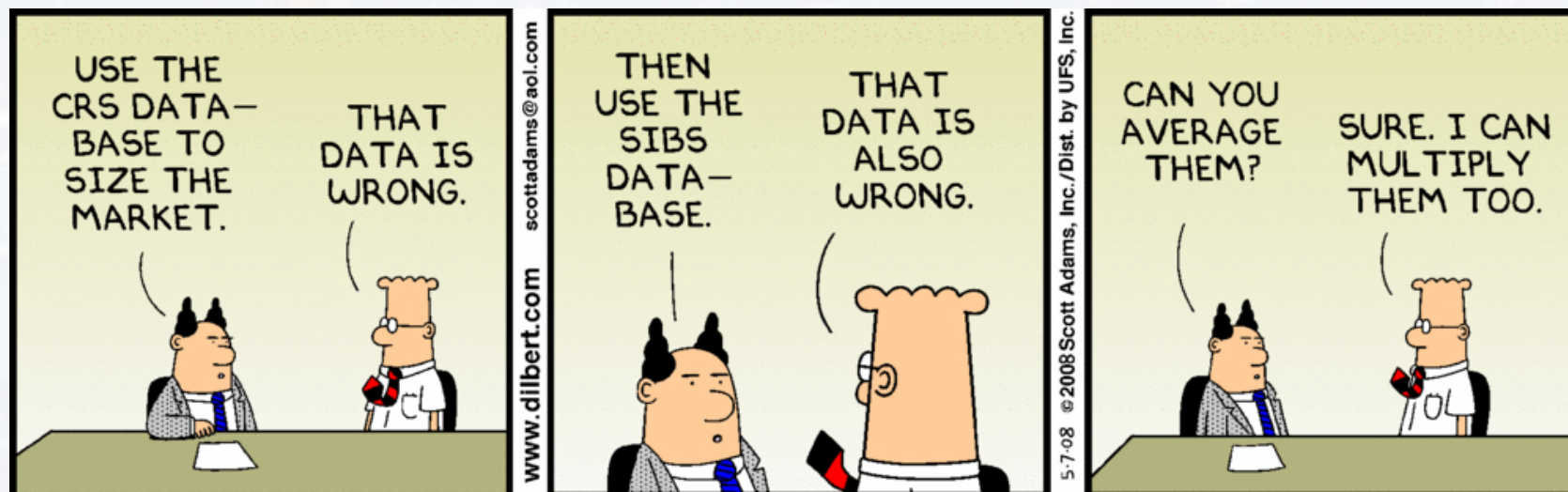
ranking tests (Wilcoxon)

value



Outline

- Expectation, Variance, and Covariance
- Hypothesis Testing
- **Confidence Intervals**

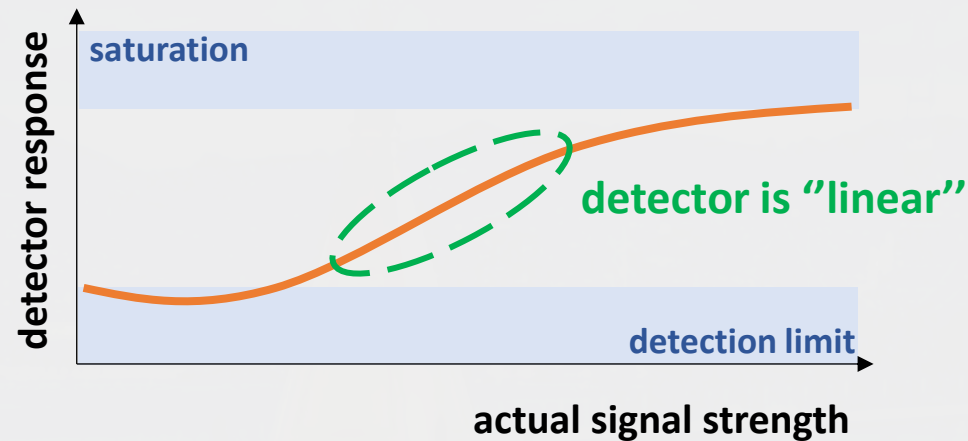




How sure can we be about a measured value?

two kinds of errors

systematic errors: calibration, non-linearity of the detector

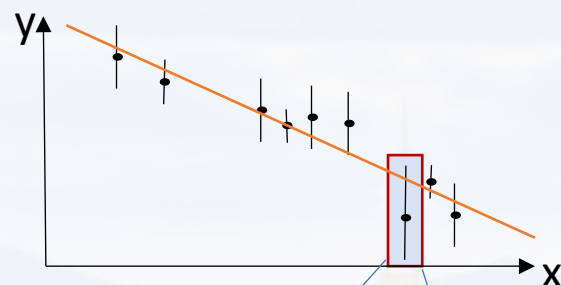


statistical errors: limited precision, natural variance of the data
→ spread of the data around **an average value**



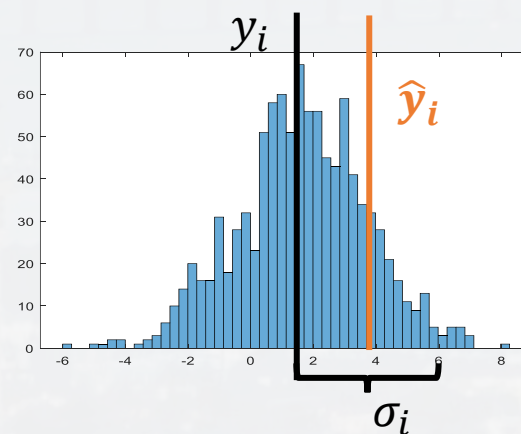
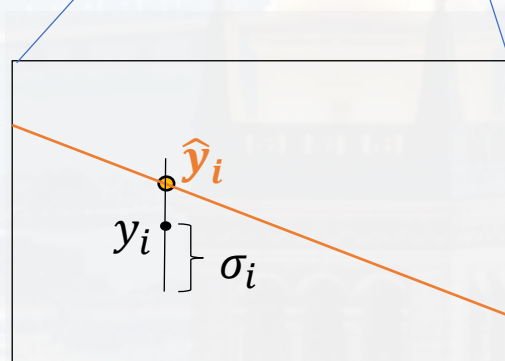
assumption: far from the detection limit and saturation:

the spread follows approximately a **normal distribution**.

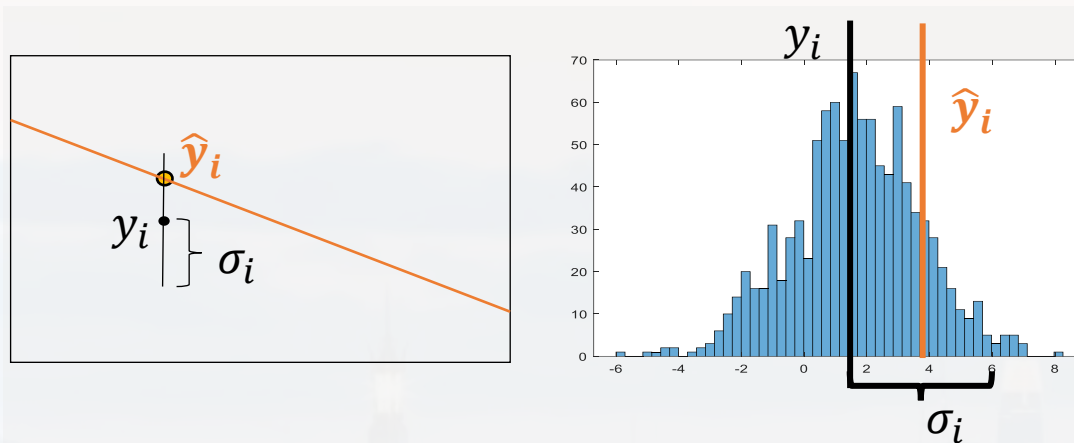


fitting a model ($\hat{y}_i$, orange line) to data points (x_i, y_i)
each with an error bar σ_i

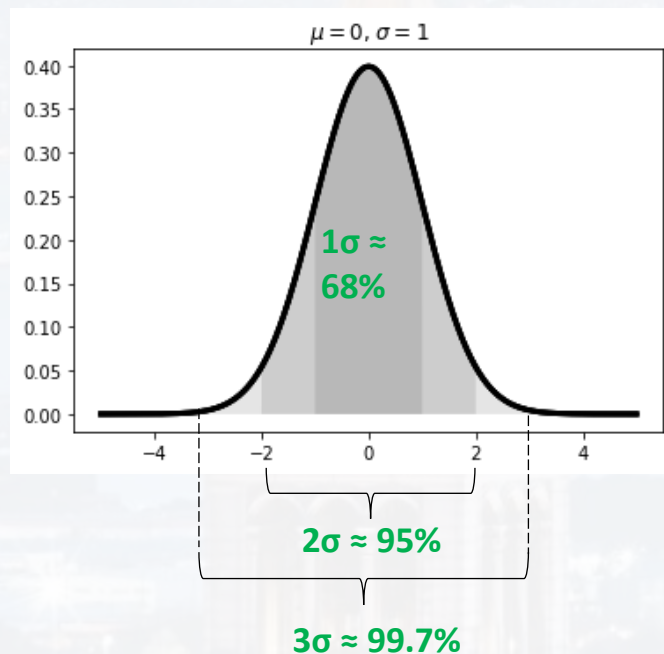
each data point x_i has been drawn from $N(\mu_i = y_i, \sigma_i)$



$$p_i(y_i|\hat{y}_i, \sigma_i) = \frac{1}{\sqrt{2\pi}\sigma_i} \exp\left[-\frac{1}{2} \frac{(y_i - \hat{y}_i)^2}{\sigma_i^2}\right]$$



$$p_i(y_i|\hat{y}_i, \sigma_i) = \frac{1}{\sqrt{2\pi}\sigma_i} \exp\left[-\frac{1}{2} \frac{(y_i - \hat{y}_i)^2}{\sigma_i^2}\right]$$



for large ($> 50 \dots 100$) N (number of data points):

$\approx 2/3$ of the data points should be consistent with the model within their 1σ error bars

$\approx 95\%$ of the data points should be consistent with the model within their 2σ error bars

$\approx 99.7\%$ of the data points should be consistent with the model within their 3σ error bars



$$p_i(y_i|\hat{y}_i, \sigma_i) = \frac{1}{\sqrt{2\pi}\sigma_i} \exp\left[-\frac{1}{2} \frac{(y_i - \hat{y}_i)^2}{\sigma_i^2}\right]$$

based on this model: reduced chi square

$$\chi_{red}^2 = \frac{1}{df} \sum_{i=1}^N \left(\frac{y_i - \hat{y}_i}{\sigma_i} \right)^2$$

$$df = N - p - 1$$

N : number of data points
p: number of fit parameter (model)

given a fitted model: χ_{red}^2 is a **measure of the fit quality!**



$$p_i(y_i|\hat{y}_i, \sigma_i) = \frac{1}{\sqrt{2\pi}\sigma_i} \exp\left[-\frac{1}{2} \frac{(y_i - \hat{y}_i)^2}{\sigma_i^2}\right]$$

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$$df = N - p - 1$$

N : number of data points
p: number of fit parameter (model)

given a fitted model: χ_{red}^2 is a **measure of the fit quality!**

example

good fit: $y_i - \hat{y}_i$ should be within σ_i for 2/3 of all data points, see $N(\mu_i = y_i, \sigma_i)$

therefore $\frac{y_i - \hat{y}_i}{\sigma_i} \approx 1$

therefore $\sum_{i=1}^N \left(\frac{y_i - \hat{y}_i}{\sigma_i} \right)^2 \approx N$

N should be **much larger** than p, therefore $df \approx N$

hence, $\chi_{red}^2 \approx 1$



$$p_i(y_i|\hat{y}_i, \sigma_i) = \frac{1}{\sqrt{2\pi}\sigma_i} \exp\left[-\frac{1}{2} \frac{(y_i - \hat{y}_i)^2}{\sigma_i^2}\right]$$

based on this model: reduced chi square

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$$p_i(y_i|\hat{y}_i, \sigma_i) = \frac{1}{\sqrt{2\pi}\sigma_i} \exp\left[-\frac{1}{2} \frac{(y_i - \hat{y}_i)^2}{\sigma_i^2}\right]$$

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$\chi_{red}^2 \approx$

1.0 excellent fit

1.0...1.5 acceptable fit

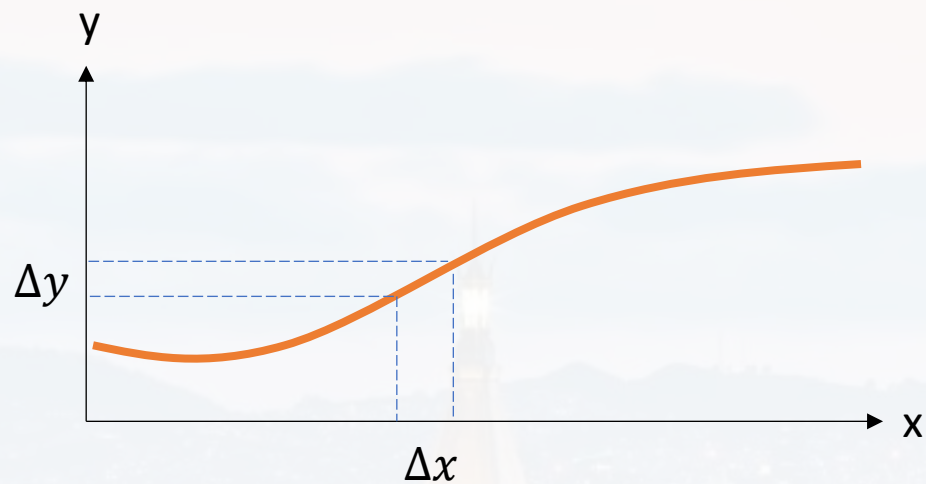
1.5...1.7 bad fit

>2.0 not acceptable

<<1.0 suspicious, errors are overestimated!



error propagation



$$\frac{dy}{dx} \approx \frac{\Delta y}{\Delta x}$$

for $\Delta x \ll x$

$$\Delta x \left| \frac{dy}{dx} \right| \approx \Delta y$$

example:

$$V = \frac{4}{3} \pi r^3$$

$\Delta V = ?$ given Δr

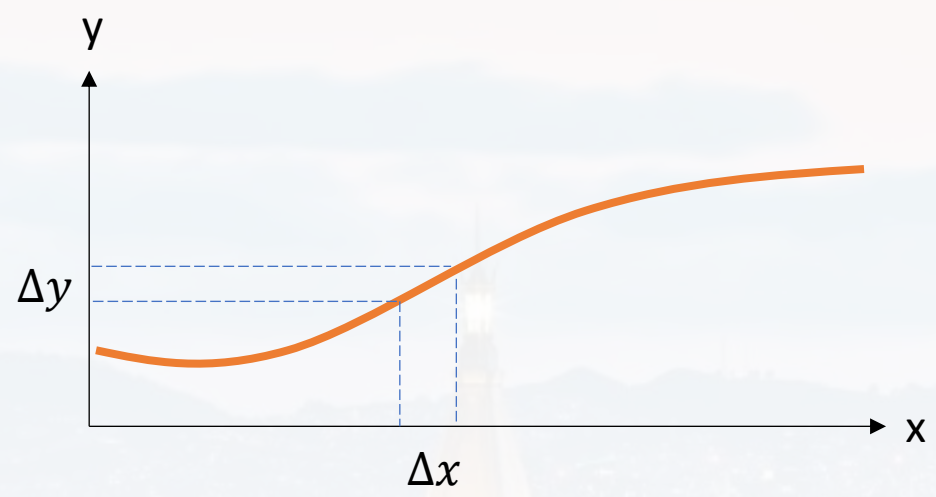
$$\Delta V = \frac{dV}{dr} \Delta r = 4 \pi r^2 \Delta r$$

$$\boxed{\frac{\Delta V}{V} = 3 \frac{\Delta r}{r}}$$

$\Delta r, \Delta V \approx 1\sigma$



error propagation



$$\frac{\Delta V}{V} = 3 \frac{\Delta r}{r}$$

$$\Delta r, \Delta V \approx 1\sigma$$

$$\Delta V = \frac{dV}{dr} \Delta r = 4 \pi r^2 \Delta r$$

radius: $r = 1.0 \mu\text{m}$
 $\Delta r = 0.1 \mu\text{m}$

$$V = \frac{4}{3} \pi (1.0 \mu\text{m})^3 = 4.18879020 \dots \mu\text{m}^3$$

$$\begin{aligned} \Delta V &= 4 \pi r^2 \Delta r = 4 \pi (1.0 \mu\text{m})^2 0.1 \mu\text{m} \\ &= 1.2566370614359172 \mu\text{m}^3 \end{aligned}$$

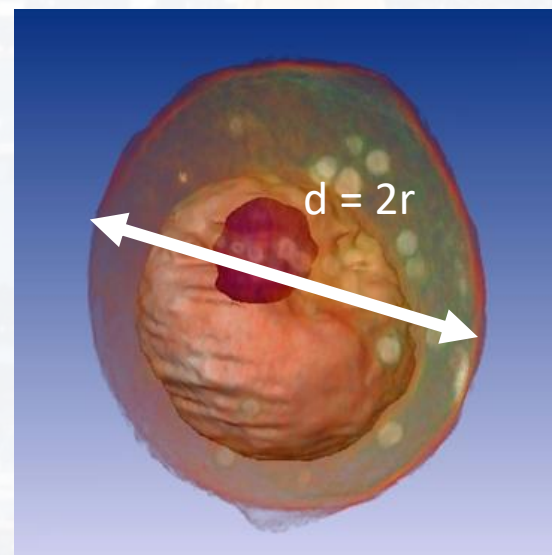
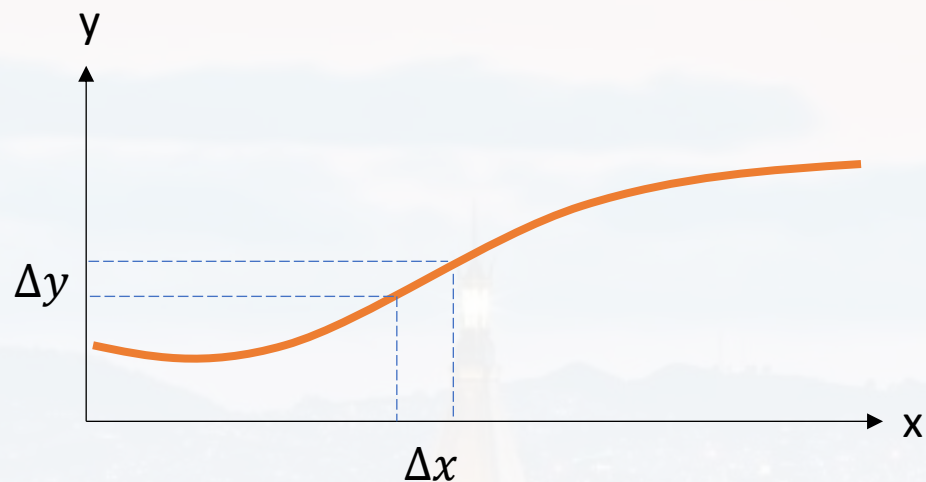


Image: NIH



error propagation



$$V = (4.2 \pm 1.3) \mu\text{m}^3$$

$$\frac{\Delta V}{V} = 3 \frac{\Delta r}{r}$$

$$\Delta r, \Delta V \approx 1\sigma$$

$$\Delta V = \frac{dV}{dr} \Delta r = 4 \pi r^2 \Delta r$$

$$\begin{array}{ll} \text{radius: } r = & 1.0 \mu\text{m} \\ \Delta r = & 0.1 \mu\text{m} \end{array}$$

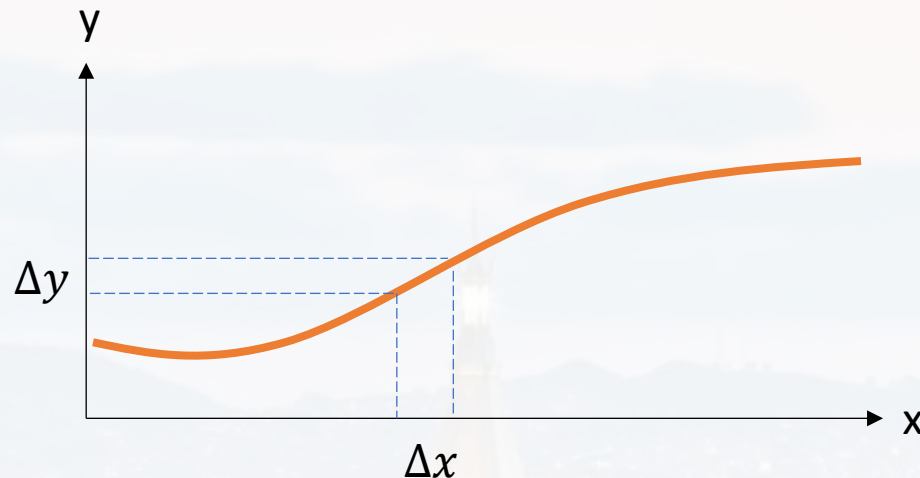
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$$\Delta V = 4 \pi r^2 \Delta r = 4 \pi (1.0 \mu\text{m})^2 0.1 \mu\text{m}$$

$$= 1.2566370614359172 \mu\text{m}^3$$



error propagation



general:

$$\Delta f(max) = \sum_{i=1}^I \left| \frac{\partial f}{\partial x_i} \right| \Delta x_i \quad \text{maximum error estimation}$$

if x_i do **not correlate**, i. e. are **mutually independent**:

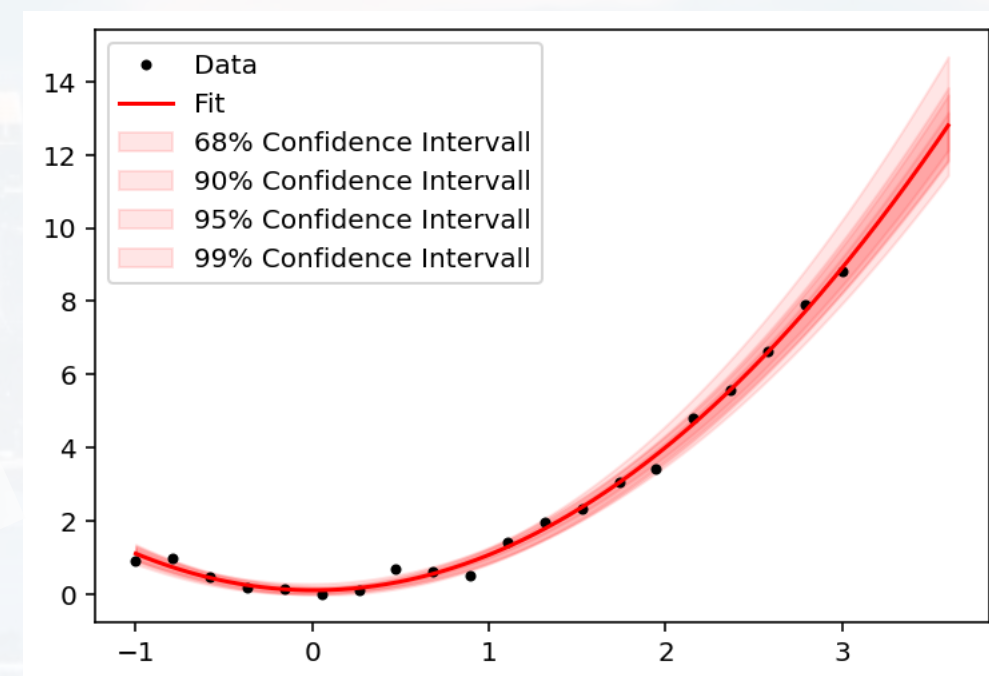
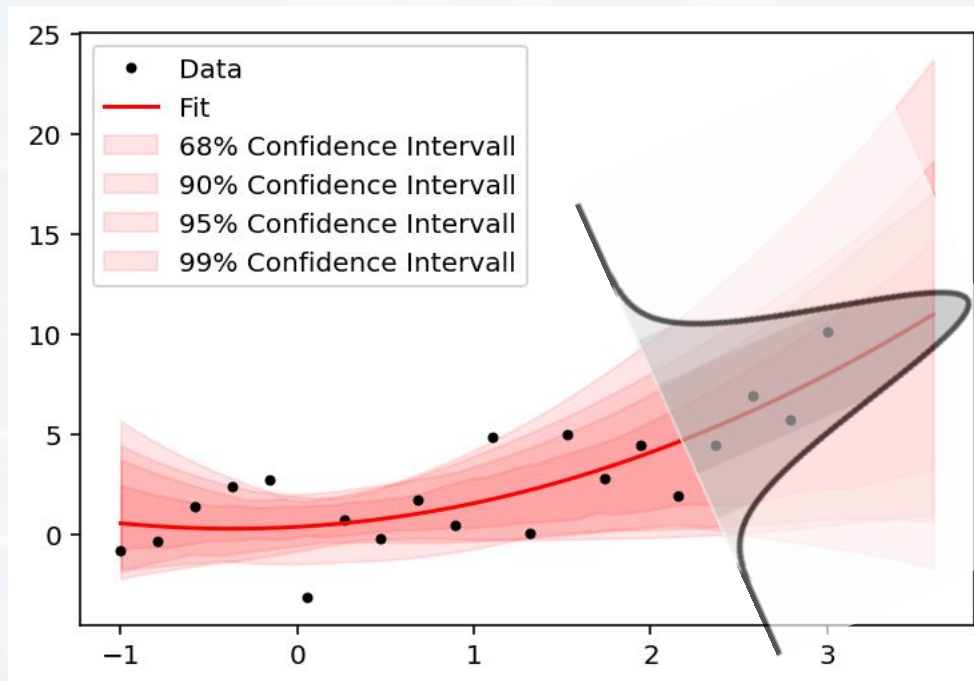
$$\Delta f^2 = \sum_{i=1}^I \left| \frac{\partial f}{\partial x_i} \right|^2 (\Delta x_i)^2$$

Note: $\Delta f(max)^2 > \Delta f^2$ because of the mixed terms $\left| \frac{\partial f}{\partial x_i} \right| \left| \frac{\partial f}{\partial x_j} \right| \Delta x_i \Delta x_j$ in $\Delta f(max)^2$



curve fitting (Module 11)

noisy data





Thank you very much for you attention!

